Sharp Inference for Longitudinal Mechanism Allocations

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Abstract

For the original finite-state, observed longitudinal graph, we give finite-sample estimation and simultaneous Euclidean confidence bounds for the complete vector of hybrid-mechanism Shapley contributions. The outcome regression is unknown and may involve overlapping sparse factors and a singular, unknown-rank design. Fixed-quota conditional tables admit an independent completion; a dimension-free bound for bounded Shapley games then controls their joint error. Conditional least squares handles the outcome using the same trajectories. Forward simulation gives polynomial randomized additive accuracy without a graph-width restriction. We prove matching regular-class rates, matching joint confidence-diameter lower bounds, and explicit exponential information examples. Every count, design, simulation and numerical failure is retained. The sharp rates use the original convention of constants independent of sample size; displayed dependence on all growing parameters is not asserted to be an exact minimax formula in every transition regime.

1 Original question, model and contribution

The canonical problem [1], motivated by the time-varying mediation agenda of [2, Section 3.4.3], specifies a finite-state longitudinal causal directed acyclic graph, T times, treatments and observed covariates, and J ordered mediators at each time. All parents are observed and each node has at most r parents. Consistency, sequential exchangeability and positivity identify the stipulated hybrid interventions. Treatments follow a fixed sequence a. Each of the $K = JT$ mediator mechanisms has active and reference kernels obtained from its observable conditional law at two specified treatment values, feasible at the generated history. The unknown bounded terminal mean is a sum of interactions of order at most r on a specified sparse graph, and its total absolute value is at most one.

For $S \subseteq [K]$, let Q_S be the resulting non-outcome trajectory law, using active mechanisms exactly on S. For a uniform permutation π, let $P_j(\pi)$ be the labels preceding j. The requested target is

$$v(S) = \mathbb{E}_{Q_S} m(H), \qquad \phi_j = \mathbb{E}_\pi[v(P_j(\pi) \cup \{j\}) - v(P_j(\pi))]. \tag{1.1}$$

Here H contains all non-outcome observations. Every permutation telescopes, so $\sum_j \phi_j = v([K]) - v(\varnothing)$. These are allocations of the joint mechanism contrast; no natural-effect or cross-world identification is substituted. The question asks for sharp estimation and simultaneous inference from n independent trajectories, polynomial graph/overlap regimes, exponential obstructions, and ordering approximation error. We use squared Euclidean loss and expected Euclidean confidence diameter. The canonical definition calls a rate sharp when its constants are independent of n for all sufficiently large n.

Finite sparse means. Supply the graph and a finite known dictionary $f = (f_1, \dots, f_p)^T$, with $|f_l(h)| \le 1$, spanning its specified local factor spaces. For example, one indicator for every cell of every prescribed finite factor scope always spans the original model, including overlapping factors. Write

$$\mathbb{E}[Y \mid H] = m(H) = f(H)^T\alpha, \qquad |m| \le 1, \qquad |Y| \le B, \quad B \ge 1. \tag{1.2}$$

The bound on m holds on observation and all required hybrid supports. The coefficients are unknown, may be nonunique and may cancel; no separate component bound or supplied outcome table is assumed. A supplied smaller bounded basis may be used when that is the declared model. We count its actual description and evaluation size. A large implicit factor table does not become small merely because it is called sparse. The uniform envelope B is a class parameter; replace an actual smaller bound by one. Section 5 also treats the union with no uniform envelope. Only the conditional mean in (1.2), not the full conditional outcome distribution, is required to be parent-local.

Rows and support. Let N count non-outcome random nodes whose conditional laws are estimated; treatment assignment tables need not be estimated for a fixed intervention. A source row is a parent address of the *same* observed node, with the stipulated source-treatment argument and the other generated parents. Tables are node- and time-specific, even if true parameters happen to be equal at different times. Distinct addresses of one node are disjoint observed parent events. Let $\mathcal{C}$ contain the rows reached by some true hybrid and suppose

$$\mathbb{P}\{\text{observed parents equal } c\} \ge \beta > 0 \quad (c \in \mathcal{C}), \qquad C = |\mathcal{C}| \le N/\beta. \tag{1.3}$$

Irrelevant structural-zero rows are allowed. The procedure need not know $\mathcal{C}$; it uses its displayed upper bound. Positivity of a treatment decision alone does not imply (1.3) for its histories.

Set $\Sigma = \mathbb{E}[f(H)f(H)^T]$ and $R = \operatorname{range}(\Sigma)$. Terminal hybrid positivity means that every generated outcome-parent state is observationally supported. In particular it implies the compatibility actually used below:

$$f(H) \in R \quad Q_S\text{-almost surely for every } S. \tag{1.4}$$

Indeed a function $f^T u$ with $u \in \ker \Sigma$ vanishes observationally on every supported outcome-parent state and hence on every hybrid state. This does not require absolute continuity of the *entire* hybrid trajectory. For example $M = A$ and a mean depending only on M can support the hybrid $A = 1, M = 0$ through its observed outcome-parent value $M = 0$. When Σ has positive rank, let

$$\lambda^+_{\min}(\Sigma) \ge \lambda > 0. \tag{1.5}$$

The rank and range are unknown. Every fixed finite positive-rank design has some such λ; quantitative uniform conclusions state it as a class bound. Rank zero is addressed separately. Thus all finite original graphs and finite r are included; r is not required to stay fixed as the graph grows.

Context and distinction from the three prior attempts. The three prior attempts [3] already prove local-table bounds, simultaneous intervals, a thin-graph rare-path lower bound, and polynomial exact calculation under bounded elimination width. Version 2 adds small mechanism-ancestor sets for supplied factors. Version 3 already learns unknown finite-dictionary coefficients under a positive-definite Gram bound and combines them with that exact calculation. Unknown-factor OLS and fixed-graph root-n exponents alone are therefore not new here. The present argument supplies a dimension-free vector bound for kernel estimation, unknown-rank identifiable quotients, matching vector-risk and joint-diameter families, and additive computation for arbitrary width, with all failure events and finite precision included. Sampling-based Shapley error bounds are classical; see [5]. We prove the particular bounds used here.

2 Estimator and complete finite-sample guarantees

If $K = 0$ or $p = 0$, the target is the zero vector and is returned exactly; the nontrivial formulas below assume $K, p \geq 1$. For a non-outcome table collection θ, define the K-by-p matrix $A(\theta)$ whose lth column is the Shapley vector with outcome function f_l. Then $\phi = A(\theta)\alpha$.

Suppose first $k = \lfloor n\beta/2 \rfloor \geq 1$. At each observed parent address retain its first k responses in a fixed trajectory order. Use that empirical conditional law when the quota is met and a fixed valid default otherwise. Store only observed addresses and look up rows lazily. Denote the fitted tables by $\widehat{\theta}$ and their analytical Shapley matrix by $\widehat{A} = A(\widehat{\theta})$; the algorithm will not compute this matrix exactly.

Let $X_{il} = f_l(H_i)$, $G = X^T X$, and P_G be the orthogonal projector onto the empirical range. Use the safe inverse

$$D = \begin{cases} G^{\dagger}, & G - (n\lambda/2)P_G \succeq 0, \\ 0, & \text{otherwise}, \end{cases} \qquad \widehat{\alpha} = DX^T Y, \qquad \widehat{b} = \|\widehat{\alpha}\|_1. \tag{2.1}$$

The symbol † is the Moore–Penrose inverse. The test retains the whole empirical inverse only when every positive empirical eigenvalue meets the threshold. It is not a test of population rank.

For each of M independent repetitions draw a uniform permutation. Simulate its $K + 1$ prefix hybrids forward under the fitted tables and evaluate $\widehat{m} = f^T\widehat{\alpha}$ on each. Assign successive differences to the switched labels, reusing each computed prefix value in adjacent differences. Average the vectors to obtain $\widehat{\phi}_{MC}$. Let z be its Euclidean projection onto the ball $\mathcal{B}_5$ of radius five centered at zero. This projection is useful for risk control; the raw vector may also be reported.

Put

$$\begin{aligned} \delta_C &= \min\{1, (N/\beta)e^{-n\beta/8}\}, & \delta_G &= \min\{1, pe^{-n\lambda/(8p)}\}, \\ \delta &= \delta_C + \delta_G, & L(a) &= 1 + \sqrt{2\log(1/a)}. \end{aligned} \tag{2.2}$$

Theorem 2.1 (All-sample risk and honest simultaneous inference)**.** *For every finite graph and unknown mean described in Section 1, with positive design rank and $k \geq 1$, the ideal finite-index implementation satisfies*

$$\mathbb{E}\|z - \phi\|_2^2 \leq \frac{25}{k} + \frac{50B^2p}{n\lambda} + \frac{8B^2Kp}{\lambda M} + 100\delta. \tag{2.3}$$

For any $a_C, a_Y, a_M \in (0, 1)$ such that $\delta + a_C + a_Y + a_M \leq a < 1$, an honest joint confidence region is

$$\mathcal{B}_5 \cap \{u : \|u - z\|_2 \leq R_n\}, \qquad R_n = \frac{5L(a_C)}{\sqrt{k}} + B\sqrt{\frac{50p}{n\lambda}}L(a_Y) + 2\widehat{b}\sqrt{\frac{K}{M}}L(a_M). \tag{2.4}$$

Its coverage is at least $1 - a$ for every n, and its diameter is at most $\min(10, 2R_n)$*. If the deterministic budgets admit no positive split, use the full ball $\mathcal{B}_5$. If $k = 0$, use the zero point and that ball. If the model has zero feature rank, its target and point estimator are identically zero. Empirical rank zero alone does not justify a singleton confidence region over an unknown-rank class.*

The confidence region covers the entire vector on one event; its coordinate projections are simultaneous intervals. We do not call their maximum coordinate width minimax-optimal in every growing-dimensional family. The raw simulated increments telescope to their matching simulated endpoint contrast. Radial projection generally changes that sample identity; it does not change the population identity in (1.1). Section 6 adds finite-bit and rounding errors to these exact mathematical bounds.

3 A bounded-game lemma and independent row completion

Lemma 3.1 (Euclidean Shapley bound)**.** *For any finite set game with $|v(S)| \le b$, its Shapley vector satisfies $\|\phi(v)\|_2 \le 5b$, independently of the number K of labels.*

Proof. $K = 0$ is empty and $K = 1$ has bound $2b$. For $K \ge 2$, fix $\|u\|_2 = 1$ and put $U = \sum_j u_j$. For $|S| = s$ with $0 < s < K$, the coefficient of $v(S)$ in $u^T\phi(v)$ is

$$\frac{K\sum_{j\in S} u_j - sU}{s(K-s)\binom{K}{s}}.$$

This follows by separating additions to $S\setminus\{j\}$ and subtractions at S in the permutation formula. The empty/full coefficients contribute $2|U|/K$ in absolute value. A uniform s-subset has mean sum sU/K and variance

$$\frac{s(K-s)}{K(K-1)}\left(1 - \frac{U^2}{K}\right).$$

One obtains this by expanding indicator covariances, using $\mathbb{P}(j \in S) = s/K$ and $\mathbb{P}(i, j \in S) = s(s-1)/(K(K-1))$. Cauchy–Schwarz bounds the absolute coefficient sum at size s by $\sqrt{K/(K-1)}\sqrt{1 - U^2/K}/\sqrt{s(K-s)}$. The function $[x(K-x)]^{-1/2}$ is convex on $(0, K)$: its second derivative is $[x(K-x)]^{-3/2} + 3(K-2x)^2/[4\{x(K-x)\}^{5/2}] > 0$. Midpoint comparison on $[s - 1/2, s + 1/2]$ therefore gives

$$\sum_{s=1}^{K-1} \frac{1}{\sqrt{s(K-s)}} \le \int_0^K \frac{dx}{\sqrt{x(K-x)}} = \pi.$$

With $t = U/\sqrt{K}$, the full coefficient sum is at most

$$\frac{2|t|}{\sqrt{K}} + \pi\sqrt{\frac{K}{K-1}}\sqrt{1-t^2} \le \sqrt{\frac{4}{K} + \frac{\pi^2 K}{K-1}} \le \sqrt{2 + 2\pi^2} < 5.$$

Multiply by b and take the supremum over unit u. □

We will use two elementary probability facts. If a variable has range length c, its log moment generating function, after centering, is at most $t^2c^2/8$. Indeed its tilted variance is at most $c^2/4$: for any variable in $[a, b]$, $(V - a)(b - V) \ge 0$ bounds its variance by $(b-a)^2/4$; integrate the second derivative of the log moment generating function twice. Revealing independent inputs of a function with single-input changes bounded by c_i and applying this fact to its Doob increments gives

$$\mathbb{E}e^{t(F-\mathbb{E}F)} \le e^{t^2\sum_i c_i^2/8}, \qquad \mathbb{P}\{F > \mathbb{E}F + s\} \le e^{-2s^2/\sum_i c_i^2}. \tag{3.1}$$

Also a binomial count with mean nq obeys $\mathbb{P}\{Z < nq/2\} \le e^{-nq/8}$. For example Markov applied to $e^{-(\log 2)Z}$ gives the stronger exponent $-nq(1 - \log 2)/2$, using $1 - q + qe^{-t} \le \exp(q(e^{-t} - 1))$.

Lemma 3.2 (Completion of fixed-quota tables)**.** *There are mutually independent completed k-sample arrays, each with its true source-row law, whose fitted hybrid games agree with the actual quota estimator on an event E_C satisfying $\mathbb{P}(E_C^c) \le \delta_C$. On E_C, every fitted hybrid support is contained in its true support.*

Proof. At a deficient row retain all available responses and append independent draws from its true conditional law until there are k. These extra draws are a proof coupling and are not available to the estimator. Condition on the observed earlier nodes of all trajectories in a topological ordering, as well as earlier auxiliary draws. Parent addresses and selected trajectory indices for the present node are fixed. The observed responses have independent row laws by

factorization of the observed DAG and independence across trajectories. After completion, every row has exactly k independent draws, and the joint conditional product law is independent of everything conditioned upon. Induction over nodes proves independence of all completed arrays. It does not assert independence of ordinary varying-count empirical tables.

Let E_C mean that every row in $\mathcal{C}$ meets quota. Binomial tails and a union bound give $\mathbb{P}(E_C^c) \le \min\{1, Ce^{-n\beta/8}\} \le \delta_C$; independence of counts is unnecessary. On this event, completed and actual tables coincide at all relevant rows. Their responses lie in true conditional supports. In a generated path a first departure from true hybrid support would need either an earlier unsupported parent path or an out-of-support response at a relevant row. Neither is possible. This induction proves both support containment and agreement for every hybrid, without enumerating coalitions. Defaults at irrelevant rows cannot be the first cause of a departure. □

Lemma 3.3 (Product-empirical vector bound)**.** *Using the true bounded mean m with the completed non-outcome arrays gives a vector $\widetilde{\phi}$ such that*

$$\mathbb{E}\|\widetilde{\phi} - \phi\|_2^2 \le 25/k, \qquad \mathbb{P}\{\|\widetilde{\phi} - \phi\|_2 > 5L(a)/\sqrt{k}\} \le a. \tag{3.2}$$

On E_C, $\widetilde{\phi} = \widehat{A}\alpha$.

Proof. Let U contain one independent draw from each relevant true row. For each S, generate a path $H_S(U)$ by reading the appropriate draw at each node. Each node-specific row is read at most once along a path. Thus $\mathbb{E}m(H_S(U)) = v(S)$. Sharing U between different hybrids is only a coupling of already identified marginals, not a causal assumption about one subject's incompatible counterfactuals. The pathwise game is bounded by one on its true supports. Its Shapley vector $g(U)$ has norm at most five by Lemma 3.1, and $\mathbb{E}g(U) = \phi$.

The completed plug-in vector is the product-empirical average of g, choosing one of the k entries independently in every row. Independently permute the k entries in each row, match entries into k diagonal tuples, and average g on those tuples. Conditional on the arrays, the expectation of this diagonal average equals the product-empirical vector. Unconditionally the k tuples are independent with the true product law, even after fixing their permutations. Jensen and the variance of an iid average give the squared bound in (3.2).

For that iid average the expected error norm is at most $5/\sqrt{k}$, and replacing one tuple changes it by at most $10/k$. Equation (3.1) bounds its exponential norm moment. Conditional Jensen for $x \mapsto \exp(t\|x\|_2)$ transfers the same upper bound to the product-empirical average. Markov optimization gives the second bound in (3.2). The last assertion follows from Lemma 3.2. No dimension or output-coordinate union appears in this argument. □

4 Unknown regression on the identifiable feature quotient

For $u \in \ker \Sigma$, $\mathbb{E}(f^T u)^2 = 0$. Taking a finite basis of this kernel proves $f \in R$ observationally. Compatibility (1.4) does the same for true hybrids. Hence $A \ker \Sigma = 0$. Fitted support containment on E_C gives

$$\widehat{A} \ker \Sigma = 0 \quad \text{on } E_C. \tag{4.1}$$

Thus coefficient nonuniqueness is harmless for both targets on that event.

We import precisely the following matrix Chernoff inequality from [4, Theorem 1.1]: independent positive semidefinite d-dimensional matrices V_i with $\|V_i\| \le R_0$ and $\mu = \lambda_{\min}(\sum_i \mathbb{E}V_i)$ satisfy

$$\mathbb{P}\left\{\lambda_{\min}\left(\sum_i V_i\right) \le (1-t)\mu\right\} \le d\left[\frac{e^{-t}}{(1-t)^{1-t}}\right]^{\mu/R_0}, \quad 0 \le t \le 1.$$

Restrict $f(H_i)f(H_i)^T$ to the fixed population space R. These matrices are independent, have norm at most p, and their expected sum has minimum eigenvalue at least $n\lambda$ there. At $t = 1/2$ the bracket is at most $e^{-1/8}$. Consequently, with probability at least $1 - \delta_G$, the event E_G holds that $G|_R \succeq (n\lambda/2)I_R$. Every sampled feature already lies in R, so on E_G, range$(G) = R$ and (2.1) uses the pseudoinverse on exactly that space. The method does not know R.

Condition on all non-outcome data, denoted $\mathcal{D}_H$. The errors $\epsilon_i = Y_i - f(H_i)^T\alpha$ are conditionally independent and centered, with variance at most B^2 and range length at most $2B$. Their absolute values need not be at most B. On $E = E_C \cap E_G$,

$$\widehat{A}(\widehat{\alpha} - \alpha) = \sum_i w_i\epsilon_i, \qquad w_i = \widehat{A}G^\dagger f(H_i), \qquad V = \sum_i \|w_i\|_2^2 = \mathrm{tr}(\widehat{A}G^\dagger\widehat{A}^T). \tag{4.2}$$

The equality uses $G^\dagger GG^\dagger = G^\dagger$. The exact conditional squared error is $\sum_i \mathrm{Var}(Y_i \mid H_i)\|w_i\|_2^2$, at most B^2V. Every column of $\widehat{A}$ is a bounded-feature game, so Lemma 3.1 implies $\|\widehat{A}\|_F^2 \le 25p$. On E_G this gives

$$V \le \frac{50p}{n\lambda}. \tag{4.3}$$

Conditional mean norm is at most $B\sqrt{V}$. A change of Y_i changes the noise norm by at most $2B\|w_i\|_2$. Thus (3.1) gives a conditional error radius $B\sqrt{V}L(a_Y)$, with failure at most a_Y. The deterministic bound (4.3) avoids exact computation of $\widehat{A}$. It is an upper bound, not a claim that every poorly conditioned feature direction affects this target.

The safe inverse also ensures on *every* sample that

$$\|\widehat{\alpha}\|_2 \le B\sqrt{2/\lambda}, \qquad |\widehat{m}(h)| \le \widehat{b} \le B\sqrt{2p/\lambda}. \tag{4.4}$$

Indeed every accepted nonzero singular value of X is at least $\sqrt{n\lambda/2}$, while $\|Y\|_2 \le B\sqrt{n}$; rejection gives zero. This controls even fallback histories. No global projection of the sparse regression function onto a difficult constraint set is used.

Proof of Theorem 2.1. Conditional on all data, each simulated increment vector W has mean $\widehat{A}\widehat{\alpha}$ and norm at most $2\widehat{b}\sqrt{K}$. Correctness follows by forward generation of each prefix and then averaging the uniform permutation. Prefix simulations may be independent or share a correctly implemented random coupling. Independent repetitions give conditional variance at most $4\widehat{b}^2K/M$ for their average. Its mean error norm is at most $2\widehat{b}\sqrt{K/M}$ and replacing one repetition changes that norm by at most $4\widehat{b}\sqrt{K}/M$. Equation (3.1) gives the last radius in (2.4).

On E, decompose the raw error into

$$(\widehat{A}\alpha - \phi) + \widehat{A}(\widehat{\alpha} - \alpha) + (\widehat{\phi}_{MC} - \widehat{A}\widehat{\alpha}).$$

The first term is a function of $\mathcal{D}_H$ and equals the completed error there. The second is centered conditional on $\mathcal{D}_H$; E also depends only on $\mathcal{D}_H$. The last is centered conditional on all data. Therefore all cross terms vanish after multiplication by $\mathbf{1}_E$ and expectation. Lemma 3.3, (4.3) and (4.4) bound the three squared terms by $25/k$, $50B^2p/(n\lambda)$ and $8B^2Kp/(\lambda M)$. Projection onto a convex set containing ϕ cannot increase distance to ϕ. On E^c both projected point and target have norm at most five, so squared error is at most 100. This proves (2.3).

The three norm tails and $\mathbb{P}(E^c) \le \delta$ give (2.4) by the triangle and union bounds. Projection again cannot increase distance to the true target. If $\Sigma = 0$, observational and hybrid features vanish, $m = 0$, and every computed coefficient is zero; this proves the rank-zero assertion. An unknown positive population direction can be absent from the sample, which explains the retained δ_G. □

Corollary 4.1 (Local outcome tables without irrelevant-cell coverage)**.** *If m is an unknown table of its observed parents, let $\beta_Y > 0$ lower-bound only parent cells used by a true hybrid. Use ordinary cell means and zero for absent cells. Then* (2.3) *improves to*

$$\frac{25}{k} + \frac{50B^2}{n\beta_Y} + \frac{4B^2K}{M} + 100(\delta_C + \delta_Y), \qquad \delta_Y = \min\{1, \beta_Y^{-1} e^{-n\beta_Y/8}\}. \tag{4.5}$$

In (2.4) *use $50/(n\beta_Y)$ instead of $50p/(n\lambda)$, B instead of $\widehat{b}$, and δ_Y instead of δ_G.*

Proof. Use disjoint cell indicators as the dictionary. For independent signs s_l, the function $\sum_l s_l f_l$ is bounded by one. Thus $\|\widehat{A}\|_F^2 = \mathbb{E}_s\|\widehat{A}s\|_2^2 \le 25$. There are at most $1/\beta_Y$ relevant cells. Except on probability δ_Y, each has count at least $n\beta_Y/2$. On E_C every fitted hybrid uses only these cells, so all other columns of $\widehat{A}$ vanish. Conditional cell-mean centering and (4.2) then give $V = \sum_c \|\widehat{A}(:,c)\|_2^2/N_c \le 50/(n\beta_Y)$. Irrelevant rare cells, including unobserved ones, contribute zero and cannot force a global inverse rejection. Every cell mean is bounded by B, so the simulation bound is $4B^2K/M$. The preceding proof applies. □

5 Sharp risk and simultaneous confidence diameter

For clarity, an honest region has a measurable membership graph and covers every member of its stated class with probability at least $1 - a$. Randomized procedures are allowed; adjoining their independent seed changes neither KL divergence nor total variation. We will repeatedly use

$$\mathrm{KL}(\mathrm{Ber}(u), \mathrm{Ber}(v)) \le \frac{(u-v)^2}{v(1-v)}, \qquad \mathrm{TV}(P,Q) \le \sqrt{\mathrm{KL}(P,Q)/2}. \tag{5.1}$$

The first inequality follows from $\log x \le x - 1$. For the second, apply the log-sum inequality to an event attaining the total variation and its complement. The resulting binary divergence is at least twice the squared difference of the event probabilities, since its second derivative in the first argument is $1/[u(1-u)] \ge 4$ and its value and first derivative vanish at equality. The log-sum inequality itself is Jensen for $x \log x$. These arguments also follow by limits at zero masses.

Proposition 5.1 (Source coverage and fixed-class sharpness)**.** *There is a binary indegree-one family within the original problem with minimax squared risk at least $c \min\{1, (n\beta)^{-1}\}$ and honest expected joint diameter at least $c_a \min\{1, (n\beta)^{-1/2}\}$ for every fixed $a \in (0,1)$. Consequently Theorem 2.1 gives sharp n^{-1} risk and $n^{-1/2}$ diameter over fixed quantitative classes containing this nondegenerate family. These lower bounds are not assertions for every graph.*

Proof. Let $A \sim \mathrm{Ber}(\beta)$, $0 < \beta \le 1/4$, $M \mid A = 0 \sim \mathrm{Ber}(1/2)$, $M \mid A = 1 \sim \mathrm{Ber}(1/2 + t)$, $|t| \le 1/4$, and $Y = M$. Treatments are randomized, both source rows and both mediator values are supported, and the sole contribution is t. Additional independent dummy mechanisms may be added. The n-sample KL between t and s is at most $(16/3)n\beta(t-s)^2$, and that to $t = 0$ is at most $4n\beta t^2$. Choose $t = \pm\Delta$ with $\Delta = \min\{1, (n\beta)^{-1/2}\}/32$. Their total variation is less than $1/4$. Nearest-target testing turns squared error below Δ^2 into correct testing. The average testing error is at least $(1 - \mathrm{TV})/2 \ge 3/8$, proving the risk lower bound.

For the diameter put $h = 1 - a$ and $b = h \min\{1, (n\beta)^{-1/2}\}/16$. Every $|t| \le b$ has sample law within $h/2$ of the zero law. Honesty therefore implies under that same zero law probability at least $h/2$ of containing each point t of the target line. Fubini gives expected intersection length at least hb; a subset of a line has length no greater than its diameter. This proves the lower bound for every $a < 1$, without a restriction $a < 1/2$. For fixed finite parameters, $M \ge nK$ makes (2.3) $O(n^{-1})$ and (2.4) $O(n^{-1/2})$ in deterministic diameter, using (4.4). The failure exponentials vanish faster than $1/n$. □

More generally the same argument applies to a class containing a smooth one-dimensional family with sample KL to its center bounded by cnt^2 and a nonzero target derivative: project onto that derivative, on a small interval where the projected derivative stays bounded away from zero, and integrate along that target coordinate. This explicitly witnessed nondegeneracy is needed for a graph-specific positive lower bound. Known dummy targets, identical mechanism laws or known constant means can instead have exactly zero risk.

Theorem 5.2 (Matched sparse-regression risk and joint diameter)**.** *For each $d \geq 1$ and $0 < \kappa \leq 1/4$, there is a fixed original-model class with $K = 2d$, $p = d + 1$, $\beta = \kappa$ and $\lambda \asymp \kappa$ whose minimax squared risk and honest expected joint Euclidean diameter, when $n\kappa \geq d^2$, are respectively*

$$\asymp \frac{p}{n\lambda}, \qquad \asymp_a \sqrt{\frac{p}{n\lambda}}.$$

The statement permits known non-outcome kernels, so its lower bounds also apply when they must be learned. The diameter conclusion holds for every fixed $a \in (0, 1)$.

Proof. Take $T = 1$, $J = 2d$, $A \sim \mathrm{Ber}(\kappa)$. Conditional on $A = 1$ the mediators are independent fair bits, and conditional on $A = 0$ they are independent Bernoulli(κ) bits. The active/reference laws are those two laws. Put $Z_j = M_{2j-1} - M_{2j}$, and use dictionary $(1, Z_1, \ldots, Z_d)$. Conditional means of distinct Z_j vanish under either treatment, giving

$$\Sigma = \mathrm{diag}(1, \sigma^2 I_d), \qquad \sigma^2 = \kappa/2 + 2\kappa(1-\kappa)^2, \qquad 13\kappa/8 \leq \sigma^2 \leq 5\kappa/2. \tag{5.2}$$

Also $\sigma^2 \leq 13/32 < 1$, so $\lambda = \sigma^2$. Let the fixed coefficient class be $\|b\|_1 \leq 1/4$, and let $Y \mid H$ be Bernoulli with mean $1/2 + \sum_j b_j Z_j$. This is a sum of unary terms, bounded between $1/4$ and $3/4$. Its outcome has $2d$ parents; take the original bound $r \geq 2d$. Every hybrid configuration is supported under observed $A = 1$. All causal assumptions are realized by independent structural draws. With $\gamma = 1/2 - \kappa \geq 1/4$, direct evaluation gives

$$\phi_b = \gamma \sum_j b_j (e_{2j-1} - e_{2j}), \qquad \|\phi_b - \phi_c\|_2^2 = 2\gamma^2 \|b - c\|_2^2. \tag{5.3}$$

The non-outcome law is fixed. Equation (5.1) gives

$$\mathrm{KL}(P_b^n, P_0^n) \leq 4n\sigma^2 \|b\|_2^2 \leq 10n\kappa\|b\|_2^2, \qquad \mathrm{KL}(P_b^n, P_c^n) \leq \tfrac{16}{3} n\sigma^2 \|b - c\|_2^2. \tag{5.4}$$

Take $b_j = \Delta s_j$, $s_j \in \{-1, 1\}$, with $\Delta = 1/\lceil 32\sqrt{n\kappa} \rceil$. Since $n\kappa \geq d^2$, $d\Delta \leq 1/32$ and all alternatives lie in the same fixed coefficient class. Neighboring signs have KL at most $5/96 < 1/16$ and total variation less than $1/4$. A wrong decoded sign of the pair contrast costs at least $2(\gamma\Delta)^2$ squared error. Pairing the uniform sign alternatives along each bit yields average error probability at least $3/8$ for that bit. Summing gives risk at least $(3/4)d\gamma^2\Delta^2 \geq cd/(n\kappa)$.

For a matching upper on this known-kernel family use

$$\widehat{b}_j = \frac{1}{n\sigma^2} \sum_i Z_{ij}(Y_i - 1/2).$$

It is unbiased, and $|Y_i - 1/2| = 1/2$ gives $\mathbb{E}\|\phi_{\widehat{b}} - \phi_b\|_2^2 \leq \gamma^2 d/(2n\sigma^2)$. Markov gives an honest Euclidean ball of radius $\sqrt{\gamma^2 d/(2an\sigma^2)}$, proving the diameter upper.

For its lower let $h = 1 - a$ and first suppose $d \geq D_a = \lceil 32 \log(2)/h \rceil$. A greedy packing of the sign cube with Hamming distance greater than $d/4$ has size $m \geq e^{d/8}$. Indeed the radius-$\lfloor d/4 \rfloor$ ball occupies at most $e^{-d/8}$ of that cube: for $V \sim \mathrm{Bin}(d, 1/2)$, $\mathbb{E}e^{-(V - d/2)} = \cosh(1/2)^d \leq e^{d/8}$

and Markov proves the claim; greedy removed balls cover the cube. Use these signs with $\Delta_a = 1/\lceil 32\sqrt{n\kappa/h}\rceil$. Their target separation is at least

$$s_n = \gamma\Delta_a\sqrt{2d} \geq \frac{\sqrt{2h}}{132}\sqrt{\frac{d}{n\kappa}}.$$

For a uniform packing index, its mutual information with the data is at most the average KL to P_0^n, at most $5hd/512$ by (5.4). This follows by subtracting the nonnegative KL of the mixture law from that average. For any decoder, conditional entropy is at most $\log 2 + \mathbb{P}(\text{error}) \log m$: reveal first whether the decoder is correct and, if not, one of at most $m-1$ labels. Since $\log m \geq d/8$, every decoder has error at least

$$1 - \frac{I + \log 2}{\log m} \geq 1 - 21h/64 = a + 43h/64.$$

Decode by the first packing target in the proposed confidence region. Coverage and diameter less than s_n imply a correct answer. Therefore the average probability of diameter at least s_n is at least $h/2$, and worst expected diameter is at least $hs_n/2$.

For $1 \leq d < D_a$ use the continuum $b = te_1$ with $|t| \leq h/(16\sqrt{n\kappa})$. Its law is within $h/2$ of P_0^n by (5.4) and (5.1). The line-segment argument of Proposition 5.1 gives expected diameter at least $\sqrt{2}h^2/(64\sqrt{n\kappa})$, hence at least $\sqrt{2}h^2\sqrt{d/(n\kappa)}/(64\sqrt{D_a})$. This covers every dimension and every fixed $a < 1$. □

The population target-aligned regression trace in this family is exactly $2d\gamma^2/\sigma^2$, of order p/λ. It is not a lower bound for a target that annihilates the corresponding directions. The condition $n\kappa \geq d^2$ is material: a dense sign box must fit the bounded total-mean class. No formula $\min\{1, p/(n\lambda)\}$ for all other joint regimes is inferred. For fixed d, κ, the theorem establishes the canonical sharp n-rates. Under uniform asymptotic coverage, apply the lower proofs with $a+\eta < 1$ for all sufficiently large n; the canonical liminf–inf condition supplies that finite-sample coverage.

The unrestricted unions. If qualitative positivity is the only uniform restriction, choose β_n of order n^{-2} in Proposition 5.1. Risk and honest diameter cannot shrink uniformly. Lemma 3.1 gives the matching constant risk upper with the zero estimator and constant diameter upper with $\mathcal{B}_5$. Likewise, with fixed non-Y overlap but no uniform outcome envelope, take a mediator with active/reference probabilities $3/4, 1/4$ and an outcome on $\{-B, B\}$ with mean tM, $|t| \leq 1/4$. Its target is $t/2$, while sample KL to $t = 0$ is uniformly $O(nt^2/B^2)$ for $t \in [-1/4, 1/4]$. Choose $B/\sqrt{n}$ sufficiently large that all these laws lie within $(1-a)/2$ of the zero law. Integrating transferred coverage over this fixed target segment gives a positive constant diameter lower for every $a < 1$; two separated targets give the risk lower. Individual boundedness therefore does not supply a uniform noise bound.

6 Polynomial computation and its full error budget

The data pass stores at most nN observed address records. No Cartesian product of possible parent values is initialized. For rational data, features and λ, the inverse rule (2.1) is an exact rational algorithm. Choose a rational full-column-rank basis U of the empirical range and compute

$$P_G = U(U^TU)^{-1}U^T, \qquad G^\dagger = U(U^TGU)^{-1}U^T.$$

Both are zero in rank zero. The second identity follows by representing G on its range and checking $GG^\dagger = G^\dagger G = P_G$. An exact rational positive-semidefiniteness test for $G-(n\lambda/2)P_G$ suffices.

Symmetric elimination either pivots on a positive diagonal and reduces to its Schur complement, encounters a negative diagonal, or has a zero diagonal with a nonzero off-diagonal entry, which certifies a negative two-dimensional quadratic value. An all-zero residual is accepted. These are polynomial rational operations; determinants and minors bound their reduced bit lengths polynomially in input length. No exact irrational eigenvector encoding is assumed.

Each simulation repetition uses $K + 1$ topological traversals and dictionary evaluations. It draws only from stored empirical responses or explicit defaults. Uniform permutations use Fisher–Yates; uniform finite indices can be drawn by rejection from fair bits in expected polynomial time. Thus runtime is polynomial in n, M, K, N, p, graph/address and dictionary description/evaluation length, and data bit length. In particular no elimination-width or mechanism-ancestry restriction is needed.

For a strict worst-case bit budget, let D_0 bound the number of finite uniform choices and $s_0 = \max(n, K, 1)$ bound their sizes. Replace each choice from s possibilities by $\lfloor sU/2^b \rfloor$ for a fresh uniform b-bit integer U. Its total variation distance from uniform is at most $s/2^b$. Stepwise maximal coupling, also for adaptive row queries, couples the entire algorithm to the ideal one except on probability at most $D_0 s_0/2^b$. Choose b to make this at most η_{bit}. Add η_{bit} to the confidence failure and $100\eta_{\mathrm{bit}}$ to the projected risk. Fixed deterministic defaults need no extra draws.

Radial projection of a rational vector uses a single square root. Bisection with rational upper bounds on its norm gives an inward rational projection within Euclidean error $\rho > 0$, in polynomial bit length and $\log(1/\rho)$. For a zero vector use zero. Add ρ to the radius; if R_* bounds the ideal risk with the bit allowance, the final risk is at most $(\sqrt{R_*} + \rho)^2$ by the L^2 triangle inequality. Logarithms and square roots in reported radii may similarly be rounded up with a declared error tolerance. These are numerical error budgets.

For prescribed additive error ε and failure a, it suffices to make the three terms in (2.4) small and require, for example,

$$n\beta \geq 8 \log \frac{4N}{\beta a}, \qquad n\lambda \geq 8p \log \frac{4p}{a}.$$

With a fixed positive failure split, this requires polynomial sample and simulation counts if $N, K, p, B, \beta^{-1}, \lambda^{-1}$, the input description/evaluation size, ε^{-1} and $\log(1/a)$ are polynomial in J, T. Bounded alphabets and interaction order, with a polynomial number of factor scopes, ensure polynomial explicit dictionary size. These are sufficient conditions; they are not necessary for a graph whose target ignores its difficult directions. For a fixed class, $M \geq nK$ and suitably small bit/rounding tolerances preserve the sharp statistical rates.

7 Exponential information and computation distinctions

Proposition 7.1 (Thin-graph rare paths, credited to the priors)**.** *For fixed $0 < \kappa < 1$, bounded-indegree binary longitudinal graphs can require exponentially many observations: a family has squared risk at least $c \min\{1, (n\kappa^T)^{-1}\}$ and honest diameter at least $c_a \min\{1, (n\kappa^T)^{-1/2}\}$.*

Proof. Use independent $A_t \sim \mathrm{Ber}(\kappa)$ and update $H_t = H_{t-1}A_t$ from $H_0 = 1$. Earlier mechanisms are irrelevant fair bits. The final mediator has probabilities $3/4$ under $A_T = 1$ and $1/4$ under $A_T = 0$. Set the terminal Bernoulli mean to $1/2 + tH_T M_T$, $|t| \leq 1/4$. Under all-one treatment the sole contribution is $t/2$. All nodes have at most two parents, the mean has interaction order two, and every required hybrid state is supported. Observations depend on t only on $H_T = M_T = 1$, of probability $(3/4)\kappa^T$. The KL/testing and continuum calculations in Proposition 5.1 apply with that probability. The risk construction already appears in all three prior versions [3]; it is included to complete this regime. □

Proposition 7.2 (Constant source coverage, exponential feature information)**.** *There is an original-model family with all non-Y source-parent masses at least* $1/2$ *and a directly evaluable two-feature mean for which minimax risk and honest expected joint diameter are, for every* n*,*

$$\asymp \min\{1, 2^d/n\}, \qquad \asymp_a \min\{1, \sqrt{2^d/n}\}, \qquad d \geq 2.$$

Its outcome interaction order and indegree are $d+1$*; they are not uniformly bounded in this example.*

Proof. Take $T = d+1$, $J = 1$, and independent fair treatments $A_1, \ldots, A_{d+1}$. At time $j+1$ observe Z_j generated from A_j, with success probability $1 - 1/d$ when $A_j = 1$ and $1/d$ when $A_j = 0$. This respects covariates before the next treatment. All structural draws are independent conditional on their displayed parents. The first d mediator mechanisms are dummy fair bits. The final mediator M has probabilities $3/4, 1/4$ at $A_{d+1} = 1, 0$. Observationally $Z_1, \ldots, Z_d, M$ are independent fair bits. Put

$$F = \Big(\prod_{j=1}^{d} Z_j\Big) M, \quad q = \mathbb{P}(F = 1) = 2^{-d-1}, \quad Y \mid H \sim \mathrm{Ber}(1/2 + tF), \quad |t| \leq 1/4.$$

The known basis $(1, F)$ is evaluated in $O(d)$ operations and its coefficient t is unknown. This is a subfamily of the specified finite factor model, so giving its simpler form to the learner strengthens a lower bound. Every non-Y parent address has mass $1/2$ or one. All generated outcome-parent configurations have positive observational probability. Take $r \geq d+1$, as the original finite but unrestricted parameter allows. Under all-one treatment put $r_d = (1 - 1/d)^d$. Only the last mechanism matters, giving

$$\phi_t = c_d t e_K, \qquad c_d = r_d/2, \qquad 1/8 \leq c_d < 1/(2e).$$

For the inequalities, $x \log(1 - 1/x)$ increases for $x \geq 2$, since its derivative is $\log(1 - 1/x) + 1/(x-1) > 0$; its limit is -1. The feature Gram is

$$\Sigma = \begin{pmatrix} 1 & q \\ q & q \end{pmatrix}, \qquad 7q/9 \leq \lambda_{\min}(\Sigma) \leq q.$$

Indeed its determinant is $q(1-q)$, its maximum eigenvalue lies in $[1, 1+q]$, and $q \leq 1/8$. The target-aligned trace equals $c_d^2/[q(1-q)]$, so this small eigenvalue is relevant to the target.

The sample KL to $t = 0$ is at most $4nqt^2$, and between t, s at most $(16/3)nq(t-s)^2$. Taking $t = \pm \min\{1, (nq)^{-1/2}\}/32$ proves the risk lower by testing. For the upper, $(nq)^{-1} \sum_i F_i(Y_i - 1/2)$ is unbiased for t with variance at most $1/(4nq)$. Clip it to $[-1/4, 1/4]$ and multiply by $c_d e_K$; its risk is at most the smaller of a constant and $c_d^2/(4nq)$. For the diameter lower, integrate honesty under the zero law over $|t| \leq (1-a)\min\{1, (nq)^{-1/2}\}/16$, exactly as before. The lower is at least $(1-a)^2 \min\{1, (nq)^{-1/2}\}/128$. Markov for the clipped estimator, intersected with the known target segment, gives the matching upper at every fixed a. □

Introducing observed AND accumulators into the last example would bound indegree but create rare non-Y parent cells. It would not preserve the constant-β claim. The two exponential examples exhibit distinct ways information becomes scarce.

Proposition 7.3 (A growing-graph coverage transition)**.** *With* $O(d)$ *binary nodes, indegree at most two and source-cell mass at least* $\beta \leq 1/4$*, a class has minimax risk at least*

$$\frac{1}{4}\Big[1 - \{1 - (1-2\beta)^n\}^d\Big]. \tag{7.1}$$

Proof. Let unknown bits $\theta_i \in \{0,1\}$ determine $Z_i = \theta_i A_i$, where A_i are independent Bernoulli(2β) treatments. Take an independent fair observed bit R, form $H_0 = R$, $H_i = H_{i-1}$ xor Z_i, and $W = H_d$ xor R. Use dummy earlier mechanisms and a final independent fair treatment whose mediator M has active/reference probabilities $3/4, 1/4$. Set $Y = M$ xor W. The construction can be ordered by putting Z_i, H_i before the next treatment. Each node has at most two parents and the outcome mean is $M + W - 2MW$. Under all-one treatment the sole effect is $\frac{1}{2}(-1)^{\sum_i \theta_i}$.

The mask R makes each H_{i-1} fair and independent of Z_i. Every relevant update-parent cell has mass at least β. If h unknown bits equal one, $\mathbb{P}(W = 1) = [1 - (1 - 4\beta)^h]/2$ is either zero or at least 2β; R is independent of W, so parent cells of the final unmasking, and then of Y, have positive mass at least β. Other required rows have mass at least β as well. Structural-zero irrelevant cells are allowed and every hybrid history is supported by the observational all-one treatment sequence.

Give the θ_i independent fair prior bits. Bit i is revealed exactly when some observed $A_i = 1$. If all its n assignments are zero, the bit remains conditionally fair: the other observed variables are functions of the observed Z_i, R, M and add no information about it. If any bit is unseen, the target sign is fair and its conditional variance is $1/4$. Missing-bit events are independent with probability $(1 - 2\beta)^n$. Bayes risk, and hence minimax risk, gives (7.1). Thus $n\beta$ substantially below $\log d$ can force constant risk; not every count failure term is a union-bound artifact. No exact formula for the whole transition window is claimed. □

Ordering error. The K/M term for naive independent-ordering averages cannot generally be removed. Take $K \geq 2$ independent fair treatments and mediators with reference success $\varepsilon = 1/(4K^2)$ and active success $1 - \varepsilon$. Binary parity accumulators have indegree two, and let $Y = (-1)^{\text{final parity}}$. All relevant non-Y parent cells have mass at least $1/4$ (some initial ones are larger). Writing $c = 1 - 2\varepsilon$, the exact game is $v(S) = c^K(-1)^{|S|}$ and $c^K \geq 1 - 1/(2K) \geq 1/2$. Each exact ordering vector has squared norm $4c^{2K}K$; its mean is zero for even K and has every coordinate $-2c^K/K$ for odd K. The exact squared error of its M-average is $4c^{2K}K/M$ for even K and $4c^{2K}(K - 1/K)/M$ for odd K. This is a statement about that sampling procedure, not about every analytic or variance-reduced method.

Exact integration versus additive accuracy. There is also a direct reduction for exact rational values. Given a Boolean circuit with G binary gates and uniform inputs, flip each gate's correct output independently with probability z. Its output probability is a polynomial of degree at most G in z: condition on the input and all gate-error bits. Exact values at $G + 1$ distinct rational $z \in [1/3, 2/3]$ allow rational interpolation at zero and recover the satisfying-assignment fraction. All gate conditional probabilities lie in $[1/3, 2/3]$. Any specified combination of at most r earlier parent bits has probability at least $(1/3)^r$, by successive conditioning and the same bound for conditional mixtures; independent fair inputs also satisfy it. Attach an independent mediator with active/reference success $3/4, 1/4$ and outcome equal to its product with the circuit output. Its single contribution is half that output probability. Consequently a polynomial exact-value oracle would solve counting satisfying assignments by polynomially many calls. This reduction is an exact-computation obstruction conditional on counting complexity; it is not an unconditional exponential-time lower bound and does not conflict with the additive simulator of Section 6.

8 Scope

Theorem 2.1 and Corollary 4.1 address every finite specified graph, all finite factor scopes and unknown coefficients, including structural zeros, singular unknown rank, arbitrary outcome parent unions and all sample sizes. Quantitative β, λ, B give explicit uniform strata; their unquantified unions have the nonshrinking conclusion in Section 5. A known zero target needs no positive lower bound. Proposition 5.1 and Theorem 5.2 prove matching regular-class rates and actual informative joint diameter, in the original n-sharp sense. Polynomial sufficient graph/overlap/input conditions, exponential information families and the complete simulation/precision budget are also given. The statistical procedure never substitutes a cross-world effect for (1.1).

The bounds do not assert a necessary-and-sufficient complexity label for every graph, an exact joint minimax formula for every varying $(n, N, p, \beta, \lambda)$, optimal maximum coordinate width, unknown graph discovery, infinite dictionaries or an unconditional exponential exact computation theorem. The original fixed finite graph and its explicit definition of a sharp n-rate are retained. These distinctions also credit the priors with the fixed-class rates implied by their own bounds.

The significant imported matrix concentration statement is specified with all its hypotheses in Section 4. The scalar concentration, testing, packing, entropy and interpolation arguments used here are given in the proof.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

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